

2016.0 RANGE ROVER (LG), 100-00

GENERAL INFORMATION

DESCRIPTION AND OPERATION

BATTERY ENERGY CONTROL MODULE (BECM) - HYBRID ELECTRIC VEHICLE

WARNINGS:

- A risk assessment must be performed before any work is undertaken.
- All hybrid electric vehicle work must be performed by a suitably qualified person.
- Appropriate Personal Protection Equipment (PPE) must be worn when working on or near a hybrid electric vehicle high voltage system.
- The hybrid electric vehicle battery pack casing must not be opened due to the risk of exposure to hazardous voltage, which may cause serious injury or death.
- The electric power inverter converter casing must not be disassembled due to the risk of exposure to hazardous voltage, which may cause serious injury or death.
- If the hybrid electric vehicle battery pack is damaged or overcharged, there is a risk of exposure to hazardous voltage and/or highly corrosive electrolyte mist. If liquid or vapour is observed leaking from the hybrid electric vehicle battery pack,

take the following action:

- Evacuate the area
- Notify manager
- Do not breathe smoke/vapour
- Contain spillage using spill kit
- Wash any spillage off body and clothing (remove contaminated clothing)
- If the hybrid electric vehicle warning indicator is illuminated, the Battery Energy Control Module (BECM) cannot isolate the high voltage cables. Live working will be necessary to rectify the fault.
- The motor generator can generate a voltage if the wheels are rotated, even if the hybrid electric vehicle system has been made safe.

CAUTION:

Diagnosis by substitution from a donor vehicle is **NOT** acceptable. Substitution of control modules does not guarantee confirmation of a fault, and may also cause additional faults in the vehicle being tested and/or the donor vehicle.

NOTES:

- If a control module or a component is suspect and the vehicle remains under manufacturer warranty, refer to the Warranty Policy and Procedures manual, or determine if any prior approval programme is in operation, prior to the installation of a new module/component.
- Generic scan tools may not read the codes listed, or may read only 5-digit codes. Match the 5 digits from the scan tool to the first 5 digits of the 7-digit code listed to identify the fault (the last 2 digits give extra information read by the manufacturer-approved diagnostic system).
- When performing voltage or resistance tests, always use a digital multimeter accurate to three decimal places, and with an up-to-date calibration certificate. When testing resistance always take the resistance of the digital multimeter leads into account.
- Check and rectify basic faults before beginning diagnostic routines involving pinpoint tests.
- Inspect connectors for signs of water ingress, and pins for damage and/or corrosion.
- If DTCs are recorded and, after performing the pinpoint tests, a fault is not present, an intermittent concern may be the cause. Always check for loose connections and corroded terminals.
- Check DDW for open campaigns. Refer to the corresponding bulletins and SSMs which may be valid for the specific customer complaint and carry out the recommendations as required.

The table below lists all Diagnostic Trouble Codes (DTCs) that could be logged in the Battery Energy Control Module (BECM) - Hybrid Electric Vehicle. For additional diagnosis and testing information, refer to the relevant Diagnosis and Testing section in the workshop manual.

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
B1206-68	Crash Occured - Event information	Event information - The battery energy control module has received a crash signal from the restraints control module	This DTC is set if the restraints control module has deployed the restraints systems following activation of the crash sensors. Check the restraints control module for DTCs and refer to the relevant DTC index
B1207-1C	Crash Input Hardwired Signal - Circuit voltage out of range	Crash signal circuit short circuit to ground, short circuit to power, open circuit, high resistance	- Refer to the electrical circuit diagrams and check the crash signal circuit for short circuit to ground, short circuit to power, open circuit, high resistance - Using the manufacturer approved diagnostic system, check the engine control module and central junction box for related DTCs and refer to the relevant DTC index
B1207-38	Crash Input Hardwired Signal - Signal frequency incorrect	Crash signal circuit short circuit to ground, short circuit to power, open circuit, high resistance	- Refer to the electrical circuit diagrams and check the crash signal circuit for short circuit to ground, short circuit to power, open circuit, high resistance - Using the manufacturer approved diagnostic system, check the engine control module and central junction box for related DTCs and refer to the relevant DTC index
B1207-39	Crash Input Hardwired Signal - Incorrect has too few pulses	Crash signal circuit short circuit to ground, short circuit to power, open circuit, high resistance	- Refer to the electrical circuit diagrams and check the crash signal circuit for short circuit to ground, short circuit to power, open circuit, high resistance - Using the manufacturer approved diagnostic system, check the engine control module and central

			junction box for related DTCs and refer to the relevant DTC index
B1207-42	Crash Input Hardwired Signal - General memory failure	- Hybrid electric vehicle battery pack failure - Battery energy control module memory failure	Using the manufacturer approved diagnostic system, clear the DTCs and retest
P0600-00	Serial Communication Link - No sub type information	Hybrid electric vehicle battery pack failure	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new hybrid electric vehicle battery pack
P0601-45	Internal Control Module Memory Checksum Error - Program memory failure	Battery energy control module is not configured correctly	- Using the manufacturer approved diagnostic system check and install the latest relevant level of software to the battery energy control module - Clear the DTCs and retest
P0601-46	Internal Control Module Memory Checksum Error - Calibration/parameter memory failure	Battery energy control module is not configured correctly	- Using the manufacturer approved diagnostic system check and install the latest relevant level of software to the battery energy control module - Clear the DTCs and retest
P06A6-00	Sensor Reference Voltage A Circuit Range/Performance - No sub type information	- Hybrid electric vehicle battery pack failure - Cell supervisor circuit reference voltage check failed	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new hybrid electric vehicle battery pack

P06A7-00	Sensor Reference Voltage B Circuit Range/Performance - No sub type information	- Hybrid electric vehicle battery pack failure - Cell supervisor circuit reference voltage check failed	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new hybrid electric vehicle battery pack
P06A8-00	Sensor Reference Voltage C Circuit Range/Performance - No sub type information	- Hybrid electric vehicle battery pack failure - Cell supervisor circuit reference voltage check failed	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new hybrid electric vehicle battery pack
P06A9-00	Sensor Reference Voltage D Circuit Range/Performance - No sub type information	- Hybrid electric vehicle battery pack failure - Cell supervisor circuit reference voltage check failed	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new hybrid electric vehicle battery pack
P06B0-00	Sensor Power Supply A Circuit/Open - No sub type information	- Hybrid electric vehicle battery pack failure - Inlet/outlet coolant temperature sensor supply voltage out of range	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new hybrid electric vehicle battery pack
P06B3-00	Sensor Power Supply B Circuit/Open - No sub type information	- Hybrid electric vehicle battery pack failure - Power supply voltage out of range for current sensor	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new hybrid electric vehicle battery pack
P06D5-	Sensor Reference	Hybrid electric	Using the manufacturer

00	Voltage "E" Circuit Range/Performance - No sub type information	vehicle battery pack failure Cell supervisor circuit reference voltage check failed	approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new hybrid electric vehicle battery pack
P06D9-00	Sensor Reference Voltage "F" Circuit Range/Performance - No sub type information	- Hybrid electric vehicle battery pack failure - Cell supervisor circuit reference voltage check failed	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new hybrid electric vehicle battery pack
P06E4-00	Control Module Wake-up Circuit Performance - No sub type information	- Hybrid electric vehicle battery pack failure - Cell supervisor circuit initialization failed	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new hybrid electric vehicle battery pack
P0A0A-13	High Voltage System Interlock Circuit - Circuit open	- High voltage connector(s) not connected correctly - Low voltage connector(s) not connected correctly - High voltage interlock loop circuit open circuit, high resistance	NOTE: The high voltage interlock loop circuit will be open circuit if any high voltage connector is not connected or a connector mechanical latch is not fully engaged - Check that all high voltage system connectors are connected correctly - Check that all low voltage connectors are connected correctly - Refer to the electrical circuit

			diagrams and check the high voltage interlock loop circuit for open circuit, high resistance
P0A0B-00	High Voltage System Interlock Circuit Performance - No sub type information	High voltage interlock loop circuit short circuit to power (primary battery)	Refer to the electrical circuit diagrams and check the high voltage interlock loop circuit for short circuit to power (primary battery)
P0A0B-13	High Voltage System Interlock Circuit Performance - Circuit open	- Hybrid electric vehicle battery pack failure - High voltage interlock loop supply open circuit	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new hybrid electric vehicle battery pack
P0A0B-1C	High Voltage System Interlock Circuit Performance - Circuit voltage out of range	- Hybrid electric vehicle battery pack failure - High voltage interlock loop supply voltage out of range	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new hybrid electric vehicle battery pack
P0A0D-12	High Voltage System Interlock Circuit High - Circuit short to battery	High voltage interlock loop circuit short circuit to power (primary battery)	Refer to the electrical circuit diagrams and check the high voltage interlock loop circuit for short circuit to power (primary battery)
P0A1F-49	Battery Energy Control Module - Internal electronic failure	- Hybrid electric vehicle battery pack failure - Cell supervisor circuit fault	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new hybrid electric vehicle battery pack
P0A7E-00	Hybrid/EV Battery Pack Over Temperature - No sub type information	- Hybrid electric vehicle battery pack failure - Cell(s) damaged - Hybrid electric	Refer to the workshop manual WARNINGS. Check for liquid or vapour being emitted from the hybrid electric vehicle battery pack vent

		vehicle battery pack overcharging - Hybrid electric vehicle battery pack cooling system fault - Air conditioning system fault	- Using the manufacturer approved diagnostic system, check the electric power inverter converter for related DTCs and refer to the relevant DTC index - Refer to the relevant section of the workshop manual and test the hybrid electric vehicle battery pack cooling system - Refer to the relevant section of the workshop manual and test the air conditioning system
P0A7F-00	Hybrid/EV Battery Pack Deterioration - No sub type information	- Hybrid electric vehicle battery pack failure - State-of-health indicates battery capability is degraded through age and usage	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new hybrid electric vehicle battery pack
P0A95-00	High Voltage Fuse - No sub type information	High voltage positive circuit short circuit to high voltage negative	NOTE: The high voltage fuse is not a serviceable part so a new hybrid electric vehicle battery pack must be installed after rectifying the circuit fault Refer to the electrical circuit diagrams and check the high voltage positive circuit for short circuit to the high voltage negative circuit. Install a new hybrid electric vehicle battery pack
P0A9B-	Hybrid/EV Battery	Hybrid electric	Using the manufacturer

00	Temperature Sensor "A" Circuit - No sub type information	vehicle battery pack failure Battery temperature sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance	approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new hybrid electric vehicle battery pack
P0A9C-00	Hybrid/EV Battery Temperature Sensor "A" Circuit Range/Performance - No sub type information	- Hybrid electric vehicle battery pack failure - Battery temperature sensor temperature measurement fault	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new hybrid electric vehicle battery pack
P0A9D-14	Hybrid/EV Battery Temperature Sensor "A" Circuit Low - Circuit short to ground or open	- Hybrid electric vehicle battery pack failure - Battery temperature sensor signal out of range low	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new hybrid electric vehicle battery pack
P0A9E-12	Hybrid/EV Battery Temperature Sensor "A" Circuit High - Circuit short to battery	- Hybrid electric vehicle battery pack failure - Battery temperature sensor signal out of range high	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new hybrid electric vehicle battery pack
P0AA1-04	Hybrid Battery Positive Contactor Circuit Stuck Closed - System internal failures	- Hybrid electric vehicle battery pack failure - Main positive contactor stuck closed	

			NOTE: This DTC indicates that the high voltage positive circuit is live and cannot be switched off. Live working will be necessary to rectify this fault Clear the DTCs and retest. If the fault persists, install a new hybrid electric vehicle battery pack
P0AA2-04	Hybrid Battery Positive Contactor Circuit Stuck Open - System internal failures	- Hybrid electric vehicle battery pack failure - Main positive contactor stuck open	Clear the DTCs and retest. If the fault persists, install a new hybrid electric vehicle battery pack
P0AA4-04	Hybrid Battery Negative Contactor Circuit Stuck Closed - System internal failures	- Hybrid electric vehicle battery pack failure - Negative contactor stuck closed	NOTE: This DTC indicates that the high voltage negative circuit is live and cannot be switched off. Live working will be necessary to rectify this fault Clear the DTCs and retest. If the fault persists, install a new hybrid electric vehicle battery pack
P0AA5-04	Hybrid Battery Negative Contactor Circuit Stuck Open - System internal failures	- Hybrid electric vehicle battery pack failure - Negative contactor stuck open	Clear the DTCs and retest. If the fault persists, install a new hybrid electric vehicle battery pack

P0AA6-04	Hybrid Battery Voltage System Isolation Fault - System internal failures	- High voltage circuit short circuit to ground - External isolation resistance below minimum limit	NOTE: Damaged high voltage cables can allow water into high voltage components - Perform a visual inspection of all HV component breather pipes (hybrid electric vehicle battery pack, electric power inverter converter and transmission) for damage and water ingress - Perform a visual inspection of all high voltage cables and connectors for damage, water ingress and corrosion - Drive the vehicle at a speed of more than 16 kph for longer than 10 minutes to allow the electric machine to dry out and the fault to clear - Clear the DTCs and retest
P0AA7-63	Hybrid Battery Voltage Isolation Sensor Circuit - Circuit/component protection time-out	- Hybrid electric vehicle battery pack ground circuit open circuit, high resistance - Hybrid electric vehicle battery pack failure - Internal or external isolation resistance measurement time-out due to vehicle conditions	- Refer to the electrical circuit diagrams and check the hybrid electric vehicle battery pack ground circuit for open circuit, high resistance - Using the manufacturer approved diagnostic system, clear the DTCs and retest

P0AA8-04	Hybrid Battery Voltage Isolation Sensor Circuit Range/Performance - System internal failures	- Hybrid electric vehicle battery pack failure - Power supply voltage out of range for isolation voltage sensor - Power supply voltage out of range for water intrusion sensor	- Refer to the electrical circuit diagrams and check the hybrid electric vehicle battery pack ground circuit for open circuit, high resistance - Using the manufacturer approved diagnostic system, clear the DTCs and retest
P0AA8-49	Hybrid Battery Voltage Isolation Sensor Circuit Range/Performance - Internal electronic failure	Hybrid electric vehicle battery pack failure	NOTE: This DTC is for information only No action required
P0AA9-49	Hybrid Battery Voltage Isolation Sensor Circuit Low - Internal electronic failure	- Hybrid electric vehicle battery pack failure - High or low voltage reference for isolation resistance measurement out of range low	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new hybrid electric vehicle battery pack
P0AAA-49	Hybrid Battery Voltage Isolation Sensor Circuit High - Internal electronic failure	- Hybrid electric vehicle battery pack failure - High or low voltage reference for isolation resistance measurement out of range high	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new hybrid electric vehicle battery pack

P0ABA-00	Hybrid Battery Pack Voltage Sense A Circuit - No sub type information	- Hybrid electric vehicle battery pack failure - Hybrid electric vehicle battery pack voltage measurement fault	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new hybrid electric vehicle battery pack
P0ABB-04	Hybrid Battery Pack Voltage Sense A Circuit Range / Performance - System internal failures	- Hybrid electric vehicle battery pack failure - Power supply voltage out of range for voltage sensor	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new hybrid electric vehicle battery pack
P0ABB-49	Hybrid Battery Pack Voltage Sense A Circuit Range / Performance - Internal electronic failure	- Hybrid electric vehicle battery pack failure - Hardware device reset or hardware diagnostics not completed for voltage sensor	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new hybrid electric vehicle battery pack
P0ABB-92	Hybrid Battery Pack Voltage Sense A Circuit Range / Performance - Performance or incorrect operation	- Hybrid electric vehicle battery pack failure - Voltage plausibility fault	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new hybrid electric vehicle battery pack
P0ABB-96	Hybrid Battery Pack Voltage Sense A Circuit Range / Performance - Component internal failure	- Hybrid electric vehicle battery pack failure - Serial peripheral interface fault for voltage sensor	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new hybrid electric vehicle battery pack
P0ABC-	Hybrid Battery Pack	Hybrid electric	Using the manufacturer

49	Voltage Sense A Circuit Low - Internal electronic failure	vehicle battery pack failure Voltage sensor signal voltage out of range low	approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new hybrid electric vehicle battery pack
P0ABD- 49	Hybrid Battery Pack Voltage Sense A Circuit High - Internal electronic failure	- Hybrid electric vehicle battery pack failure - Voltage sensor signal voltage out of range high	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new hybrid electric vehicle battery pack
P0ABF- 00	Hybrid Battery Pack Current Sensor A Circuit - No sub type information	- Hybrid electric vehicle battery pack failure - Current sensor circuit open circuit	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new hybrid electric vehicle battery pack
P0AC0- 00	Hybrid Battery Pack Current Sensor A Circuit Range / Performance - No sub type information	- Hybrid electric vehicle battery pack failure - Current plausibility fault	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new hybrid electric vehicle battery pack
P0AC0- 49	Hybrid Battery Pack Current Sensor A Circuit Range / Performance - Internal electronic failure	- Hybrid electric vehicle battery pack failure - Hardware device reset or hardware diagnostics not completed for current sensor	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new hybrid electric vehicle battery pack
P0AC0- 96	Hybrid Battery Pack Current Sensor A Circuit Range / Performance - Component internal failure	- Hybrid electric vehicle battery pack failure - Serial peripheral interface fault for current sensor	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new hybrid electric vehicle battery pack

P0AC1-49	Hybrid Battery Pack Current Sensor A Circuit Low - Internal electronic failure	- Hybrid electric vehicle battery pack failure - Current out of range low	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new hybrid electric vehicle battery pack
P0AC2-49	Hybrid Battery Pack Current Sensor A Circuit High - Internal electronic failure	- Hybrid electric vehicle battery pack failure - Current out of range high	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new hybrid electric vehicle battery pack
P0AC5-00	Hybrid Battery Temperature Sensor B Circuit - No sub type information	- Hybrid electric vehicle battery pack failure - Battery temperature sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new hybrid electric vehicle battery pack
P0AC6-00	Hybrid Battery Temperature Sensor B Circuit Range/Performance - No sub type information	- Hybrid electric vehicle battery pack failure - Battery temperature sensor temperature measurement fault	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new hybrid electric vehicle battery pack
P0AC7-14	Hybrid/EV Battery Temperature Sensor "B" Circuit Low - Circuit short to ground or open	- Hybrid electric vehicle battery pack failure - Battery temperature sensor signal out of range low	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new hybrid electric vehicle battery pack
P0AC8-12	Hybrid/EV Battery Temperature Sensor	Hybrid electric vehicle battery	Using the manufacturer approved diagnostic

	"B" Circuit High - Circuit short to battery	pack failure ▪ Battery temperature sensor signal out of range high	system, clear the DTCs and retest. If the fault persists, install a new hybrid electric vehicle battery pack
P0ACA-00	Hybrid Battery Temperature Sensor C Circuit - No sub type information	▪ Hybrid electric vehicle battery pack failure ▪ Battery temperature sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance	▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new hybrid electric vehicle battery pack
P0ACB-00	Hybrid Battery Temperature Sensor C Circuit Range/Performance - No sub type information	▪ Hybrid electric vehicle battery pack failure ▪ Battery temperature sensor temperature measurement fault	▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new hybrid electric vehicle battery pack
P0ACC-14	Hybrid/EV Battery Temperature Sensor "C" Circuit Low - Circuit short to ground or open	▪ Hybrid electric vehicle battery pack failure ▪ Battery temperature sensor signal out of range low	▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new hybrid electric vehicle battery pack
P0ACD-12	Hybrid/EV Battery Temperature Sensor "C" Circuit High - Circuit short to battery	▪ Hybrid electric vehicle battery pack failure ▪ Battery temperature sensor signal out of range high	▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new hybrid electric vehicle battery pack
P0ADA-	Hybrid Battery	▪ Hybrid electric	▪ Using the manufacturer

04	Positive Contactor Control Circuit Range/Performance - System internal failures	vehicle battery pack failure Main positive contactor driver low side circuit voltage above threshold	approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new hybrid electric vehicle battery pack
P0ADB-49	Hybrid Battery Positive Contactor Control Circuit Low - Internal electronic failure	- Hybrid electric vehicle battery pack failure - Main positive contactor driver low side circuit short circuit to ground, open circuit	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new hybrid electric vehicle battery pack
P0ADC-49	Hybrid Battery Positive Contactor Control Circuit High - Internal electronic failure	- Hybrid electric vehicle battery pack failure - Main positive contactor driver low side circuit short circuit to power	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new hybrid electric vehicle battery pack
P0ADE-00	Hybrid Battery Negative Contactor Control Circuit Range/Performance - No sub type information	- Hybrid electric vehicle battery pack failure - Negative contactor driver low side circuit short circuit to power	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new hybrid electric vehicle battery pack
P0ADF-49	Hybrid Battery Negative Contactor Control Circuit Low - Internal electronic failure	- Hybrid electric vehicle battery pack failure - Negative contactor driver low side circuit short circuit to ground, open circuit	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new hybrid electric vehicle battery pack

P0AE0-00	Hybrid Battery Negative Contactor Control Circuit High - No sub type information	- Hybrid electric vehicle battery pack failure - Negative contactor driver low side circuit voltage above threshold	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new hybrid electric vehicle battery pack
P0AE1-04	Hybrid Battery Precharge Contactor Circuit - System internal failures	- High voltage direct current bus short circuit - High voltage positive circuit short circuit to high voltage negative	NOTE: This DTC indicates that there may be a short circuit of the high voltage direct current bus, outside of the hybrid electric vehicle battery pack. It is possible that the high voltage positive circuit is live and cannot be switched off. Live working will be necessary to rectify this fault - Check the other high voltage components for DTCs indicating a high voltage direct current bus short circuit and refer to the relevant DTC index - Disconnect the hybrid electric vehicle battery pack from the high voltage direct current bus, and check the high voltage direct current bus for short circuit (high voltage positive to high voltage negative). Disconnect other high voltage components to trace the fault to a specific component - Using the manufacturer approved diagnostic

			system, clear the DTCs and retest
P0AE5-00	Hybrid Battery Precharge Contactor Control Circuit Range/Performance - No sub type information	- Hybrid electric vehicle battery pack failure - High voltage direct current bus short circuit	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new hybrid electric vehicle battery pack
P0AE6-49	Hybrid Battery Precharge Contactor Control Circuit Low - Internal electronic failure	- Hybrid electric vehicle battery pack failure - Pre-charge positive contactor driver low side circuit short circuit to ground, open circuit	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new hybrid electric vehicle battery pack
P0AE7-49	Hybrid Battery Precharge Contactor Control Circuit High - Internal electronic failure	- Hybrid electric vehicle battery pack failure - Pre-charge positive contactor driver low side circuit short circuit to power	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new hybrid electric vehicle battery pack
P0AE8-00	Hybrid Battery Temperature Sensor D Circuit - No sub type information	- Hybrid electric vehicle battery pack failure - Battery temperature sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new hybrid electric vehicle battery pack
P0AE9-00	Hybrid Battery Temperature Sensor	Hybrid electric vehicle battery	Using the manufacturer approved diagnostic

	"D" Circuit Range/Performance - No sub type information	pack failure Battery temperature sensor temperature measurement fault	system, clear the DTCs and retest. If the fault persists, install a new hybrid electric vehicle battery pack
P0AEA-14	Hybrid/EVBattery Temperature Sensor "D" Circuit Low - Circuit short to ground or open	- Hybrid electric vehicle battery pack failure - Battery temperature sensor signal out of range low	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new hybrid electric vehicle battery pack
P0AEB-12	Hybrid/EVBattery Temperature Sensor "D" Circuit High - Circuit short to battery	- Hybrid electric vehicle battery pack failure - Battery temperature sensor signal out of range high	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new hybrid electric vehicle battery pack
P0AF8-00	Hybrid Battery System Voltage - No sub type information	- Hybrid electric vehicle battery pack failure - Determination of maximum or minimum cell voltage differs between primary and secondary	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new hybrid electric vehicle battery pack
P0AF9-00	Hybrid Battery System Voltage Unstable - No sub type information	- Hybrid electric vehicle battery pack failure - Suspected cell voltage reversal	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new hybrid electric vehicle battery pack
P0AFA-00	Hybrid Battery System Voltage Low - No sub type information	- Hybrid electric vehicle battery pack failure - Suspected cell	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists,

		internal short circuit	install a new hybrid electric vehicle battery pack
P0AFB-00	Hybrid Battery System Voltage High - No sub type information	- Hybrid electric vehicle battery pack failure - Suspected over-charge due to lithium plating	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new hybrid electric vehicle battery pack
P0AFC-73	Hybrid Battery Pack Sensor Module - Actuator stuck closed	- Hybrid electric vehicle battery pack failure - Cell balancing deep discharge fault	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new hybrid electric vehicle battery pack
P0B15-04	Hybrid Battery Pack Voltage Sense B Circuit Range / Performance - System internal failures	- Hybrid electric vehicle battery pack failure - Power supply voltage out of range for voltage sensor	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new hybrid electric vehicle battery pack
P0B15-49	Hybrid Battery Pack Voltage Sense B Circuit Range / Performance - Internal electronic failure	- Hybrid electric vehicle battery pack failure - Hardware device reset or hardware diagnostics not completed for voltage sensor	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new hybrid electric vehicle battery pack
P0B15-96	Hybrid Battery Pack Voltage Sense B Circuit Range / Performance - Component internal failure	- Hybrid electric vehicle battery pack failure - Serial peripheral interface fault for voltage sensor	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new hybrid electric vehicle battery pack

P0B17-49	Hybrid Battery Pack Voltage Sense B Circuit High - Internal electronic failure	- Hybrid electric vehicle battery pack failure - Voltage sensor signal voltage out of range high	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new hybrid electric vehicle battery pack
P0B1A-04	Hybrid Battery Pack Voltage Sense C Circuit Range / Performance - System internal failures	- Hybrid electric vehicle battery pack failure - Power supply voltage out of range for voltage sensor	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new hybrid electric vehicle battery pack
P0B1A-49	Hybrid Battery Pack Voltage Sense C Circuit Range / Performance - Internal electronic failure	- Hybrid electric vehicle battery pack failure - Hardware device reset or hardware diagnostics not completed for voltage sensor	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new hybrid electric vehicle battery pack
P0B1A-96	Hybrid Battery Pack Voltage Sense C Circuit Range / Performance - Component internal failure	- Hybrid electric vehicle battery pack failure - Serial peripheral interface fault for voltage sensor	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new hybrid electric vehicle battery pack
P0B1B-49	Hybrid Battery Pack Voltage Sense C Circuit Low - Internal electronic failure	- Hybrid electric vehicle battery pack failure - Voltage sensor signal voltage out of range low	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new hybrid electric vehicle battery pack
P0B1C-49	Hybrid Battery Pack Voltage Sense C	Hybrid electric vehicle battery	Using the manufacturer approved diagnostic

	Circuit High - Internal electronic failure	pack failure ▪ Voltage sensor signal voltage out of range high	system, clear the DTCs and retest. If the fault persists, install a new hybrid electric vehicle battery pack
P0B1F-04	Hybrid Battery Pack Voltage Sense D Circuit Range / Performance - System internal failures	▪ Hybrid electric vehicle battery pack failure ▪ Power supply voltage out of range for voltage sensor	▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new hybrid electric vehicle battery pack
P0B1F-49	Hybrid Battery Pack Voltage Sense D Circuit Range / Performance - Internal electronic failure	▪ Hybrid electric vehicle battery pack failure ▪ Hardware device reset or hardware diagnostics not completed for voltage sensor	▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new hybrid electric vehicle battery pack
P0B1F-96	Hybrid Battery Pack Voltage Sense D Circuit Range / Performance - Component internal failure	▪ Hybrid electric vehicle battery pack failure ▪ Serial peripheral interface fault for voltage sensor	▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new hybrid electric vehicle battery pack
P0B20-49	Hybrid Battery Pack Voltage Sense D Circuit Low - Internal electronic failure	▪ Hybrid electric vehicle battery pack failure ▪ Voltage sensor signal voltage out of range low	▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new hybrid electric vehicle battery pack
P0B21-49	Hybrid Battery Pack Voltage Sense D Circuit High - Internal electronic failure	▪ Hybrid electric vehicle battery pack failure ▪ Voltage sensor signal	▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new hybrid electric

		voltage out of range high	vehicle battery pack
P0B3B-00	Hybrid Battery Voltage Sense A Circuit - No sub type information	- Hybrid electric vehicle battery pack failure - Cell supervisor circuit 0 fault	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new hybrid electric vehicle battery pack
P0B3C-00	Hybrid Battery Voltage Sense A Circuit Range/Performance - No sub type information	- Hybrid electric vehicle battery pack failure - Offset balance voltage accuracy check failed	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new hybrid electric vehicle battery pack
P0B3D-96	Hybrid Battery Voltage Sense A Circuit Low - Component internal failure	- Hybrid electric vehicle battery pack failure - Cell voltage out of range low	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new hybrid electric vehicle battery pack
P0B3E-96	Hybrid Battery Voltage Sense A Circuit High - Component internal failure	- Hybrid electric vehicle battery pack failure - Cell voltage out of range high	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new hybrid electric vehicle battery pack
P0B3F-00	Hybrid Battery Voltage Sense A Circuit Intermittent / Erratic - No sub type information	- Hybrid electric vehicle battery pack failure - Difference between cell supervisor circuit module voltage and cell supervisor circuit sum of cells is greater than a calibratable threshold	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new hybrid electric vehicle battery pack

P0B3F-04	Hybrid Battery Voltage Sense A Circuit Intermittent / Erratic - System internal failures	- Hybrid electric vehicle battery pack failure - Number of invalid cell voltage sensor measurements exceeds calibratable threshold	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new hybrid electric vehicle battery pack
P0B3F-49	Hybrid Battery Voltage Sense A Circuit Intermittent / Erratic - Internal electronic failure	- Hybrid electric vehicle battery pack failure - Number of invalid cell voltage sensor measurements exceeds calibratable threshold	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new hybrid electric vehicle battery pack
P0B3F-92	Hybrid Battery Voltage Sense A Circuit Intermittent / Erratic - Performance or incorrect operation	- Hybrid electric vehicle battery pack failure - Number of invalid cell voltage sensor measurements exceeds calibratable threshold	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new hybrid electric vehicle battery pack
P0B3F-96	Hybrid Battery Voltage Sense A Circuit Intermittent / Erratic - Component internal failure	- Hybrid electric vehicle battery pack failure - Number of invalid cell voltage sensor measurements exceeds calibratable threshold	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new hybrid electric vehicle battery pack
P0B40-00	Hybrid Battery Voltage Sense B	Hybrid electric vehicle battery	Using the manufacturer approved diagnostic

	Circuit - No sub type information	pack failure Cell supervisor circuit 1 fault	system, clear the DTCs and retest. If the fault persists, install a new hybrid electric vehicle battery pack
P0B41-00	Hybrid Battery Voltage Sense B Circuit Range/Performance - No sub type information	- Hybrid electric vehicle battery pack failure - Offset balance voltage accuracy check failed	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new hybrid electric vehicle battery pack
P0B42-96	Hybrid Battery Voltage Sense B Circuit Low - Component internal failure	- Hybrid electric vehicle battery pack failure - Cell voltage out of range low	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new hybrid electric vehicle battery pack
P0B43-96	Hybrid Battery Voltage Sense B Circuit High - Component internal failure	- Hybrid electric vehicle battery pack failure - Cell voltage out of range high	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new hybrid electric vehicle battery pack
P0B44-00	Hybrid Battery Voltage Sense B Circuit Intermittent / Erratic - No sub type information	- Hybrid electric vehicle battery pack failure - Difference between cell supervisor circuit module voltage and cell supervisor circuit sum of cells is greater than a calibratable threshold	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new hybrid electric vehicle battery pack
P0B45-00	Hybrid Battery Voltage Sense C Circuit - No sub type information	- Hybrid electric vehicle battery pack failure - Cell supervisor circuit 2 fault	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new hybrid electric vehicle battery pack

P0B46-00	Hybrid Battery Voltage Sense C Circuit Range/Performance - No sub type information	- Hybrid electric vehicle battery pack failure - Offset balance voltage accuracy check failed	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new hybrid electric vehicle battery pack
P0B47-96	Hybrid Battery Voltage Sense C Circuit Low - Component internal failure	- Hybrid electric vehicle battery pack failure - Cell voltage out of range low	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new hybrid electric vehicle battery pack
P0B48-96	Hybrid Battery Voltage Sense C Circuit High - Component internal failure	- Hybrid electric vehicle battery pack failure - Cell voltage out of range high	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new hybrid electric vehicle battery pack
P0B49-00	Hybrid Battery Voltage Sense C Circuit Intermittent / Erratic - No sub type information	- Hybrid electric vehicle battery pack failure - Difference between cell supervisor circuit module voltage and cell supervisor circuit sum of cells is greater than a calibratable threshold	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new hybrid electric vehicle battery pack
P0B4A-00	Hybrid Battery Voltage Sense D Circuit - No sub type information	- Hybrid electric vehicle battery pack failure - Cell supervisor circuit 3 fault	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new hybrid electric vehicle battery pack
P0B4B-00	Hybrid Battery Voltage Sense D Circuit Range/Performance -	- Hybrid electric vehicle battery pack failure - Offset balance	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists,

	No sub type information	voltage accuracy check failed	install a new hybrid electric vehicle battery pack
P0B4C-96	Hybrid Battery Voltage Sense D Circuit Low - Component internal failure	- Hybrid electric vehicle battery pack failure - Cell voltage out of range low	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new hybrid electric vehicle battery pack
P0B4D-96	Hybrid Battery Voltage Sense D Circuit High - Component internal failure	- Hybrid electric vehicle battery pack failure - Cell voltage out of range high	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new hybrid electric vehicle battery pack
P0B4E-00	Hybrid Battery Voltage Sense D Circuit Intermittent / Erratic - No sub type information	- Hybrid electric vehicle battery pack failure - Difference between cell supervisor circuit module voltage and cell supervisor circuit sum of cells is greater than a calibratable threshold	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new hybrid electric vehicle battery pack
P0B4F-00	Hybrid Battery Voltage Sense E Circuit - No sub type information	- Hybrid electric vehicle battery pack failure - Cell supervisor circuit 4 fault	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new hybrid electric vehicle battery pack
P0B50-00	Hybrid Battery Voltage Sense E Circuit Range/Performance - No sub type information	- Hybrid electric vehicle battery pack failure - Offset balance voltage accuracy check failed	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new hybrid electric vehicle battery pack

P0B51-96	Hybrid Battery Voltage Sense E Circuit Low - Component internal failure	- Hybrid electric vehicle battery pack failure - Cell voltage out of range low	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new hybrid electric vehicle battery pack
P0B52-96	Hybrid Battery Voltage Sense E Circuit High - Component internal failure	- Hybrid electric vehicle battery pack failure - Cell voltage out of range high	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new hybrid electric vehicle battery pack
P0B53-00	Hybrid Battery Voltage Sense E Circuit Intermittent / Erratic - No sub type information	- Hybrid electric vehicle battery pack failure - Difference between cell supervisor circuit module voltage and cell supervisor circuit sum of cells is greater than a calibratable threshold	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new hybrid electric vehicle battery pack
P0B54-00	Hybrid Battery Voltage Sense F Circuit - No sub type information	- Hybrid electric vehicle battery pack failure - Cell supervisor circuit 5 fault	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new hybrid electric vehicle battery pack
P0B55-00	Hybrid Battery Voltage Sense F Circuit Range/Performance - No sub type information	- Hybrid electric vehicle battery pack failure - Offset balance voltage accuracy check failed	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new hybrid electric vehicle battery pack
P0B56-96	Hybrid Battery Voltage Sense F Circuit Low - Component internal failure	- Hybrid electric vehicle battery pack failure - Cell voltage out of range	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new hybrid electric

		low	vehicle battery pack
P0B57-96	Hybrid Battery Voltage Sense F Circuit High - Component internal failure	- Hybrid electric vehicle battery pack failure - Cell voltage out of range high	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new hybrid electric vehicle battery pack
P0B58-00	Hybrid Battery Voltage Sense F Circuit Intermittent / Erratic - No sub type information	- Hybrid electric vehicle battery pack failure - Difference between cell supervisor circuit module voltage and cell supervisor circuit sum of cells is greater than a calibratable threshold	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new hybrid electric vehicle battery pack
P0B5B-96	Hybrid Battery Voltage Sense G Circuit Low - Component internal failure	- Hybrid electric vehicle battery pack failure - Cell voltage out of range low	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new hybrid electric vehicle battery pack
P0B5C-96	Hybrid Battery Voltage Sense G Circuit High - Component internal failure	- Hybrid electric vehicle battery pack failure - Cell voltage out of range high	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new hybrid electric vehicle battery pack
P0B60-96	Hybrid Battery Voltage Sense H Circuit Low - Component internal failure	- Hybrid electric vehicle battery pack failure - Cell voltage out of range low	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new hybrid electric vehicle battery pack
P0B61-96	Hybrid Battery Voltage Sense H Circuit High -	Hybrid electric vehicle battery pack failure	Using the manufacturer approved diagnostic system, clear the DTCs and

	Component internal failure	Cell voltage out of range high	retest. If the fault persists, install a new hybrid electric vehicle battery pack
P0B65-96	Hybrid Battery Voltage Sense I Circuit Low - Component internal failure	- Hybrid electric vehicle battery pack failure - Cell voltage out of range low	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new hybrid electric vehicle battery pack
P0B66-96	Hybrid Battery Voltage Sense I Circuit High - Component internal failure	- Hybrid electric vehicle battery pack failure - Cell voltage out of range high	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new hybrid electric vehicle battery pack
P0B6A-96	Hybrid Battery Voltage Sense J Circuit Low - Component internal failure	- Hybrid electric vehicle battery pack failure - Cell voltage out of range low	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new hybrid electric vehicle battery pack
P0B6B-96	Hybrid Battery Voltage Sense J Circuit High - Component internal failure	- Hybrid electric vehicle battery pack failure - Cell voltage out of range high	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new hybrid electric vehicle battery pack
P0B6F-96	Hybrid Battery Voltage Sense K Circuit Low - Component internal failure	- Hybrid electric vehicle battery pack failure - Cell voltage out of range low	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new hybrid electric vehicle battery pack
P0B70-96	Hybrid Battery Voltage Sense K Circuit High - Component internal failure	- Hybrid electric vehicle battery pack failure - Cell voltage out of range high	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new hybrid electric vehicle battery pack

P0B74-96	Hybrid Battery Voltage Sense L Circuit Low - Component internal failure	- Hybrid electric vehicle battery pack failure - Cell voltage out of range low	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new hybrid electric vehicle battery pack
P0B75-96	Hybrid Battery Voltage Sense L Circuit High - Component internal failure	- Hybrid electric vehicle battery pack failure - Cell voltage out of range high	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new hybrid electric vehicle battery pack
P0BC2-00	Hybrid Battery Temperature Sensor E Circuit - No sub type information	- Hybrid electric vehicle battery pack failure - Battery temperature sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new hybrid electric vehicle battery pack
P0BC3-00	Hybrid Battery Temperature Sensor "E" Circuit Range/Performance - No sub type information	- Hybrid electric vehicle battery pack failure - Battery temperature sensor temperature measurement fault	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new hybrid electric vehicle battery pack
P0BC4-14	Hybrid/EV Battery Temperature Sensor E Circuit Low - Circuit short to ground or open	- Hybrid electric vehicle battery pack failure - Battery temperature sensor signal out of range low	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new hybrid electric vehicle battery pack
P0BC5-12	Hybrid/EV Battery Temperature Sensor E Circuit High - Circuit	Hybrid electric vehicle battery pack failure	Using the manufacturer approved diagnostic system, clear the DTCs and

	short to battery	▪ Battery temperature sensor signal out of range high	retest. If the fault persists, install a new hybrid electric vehicle battery pack
P0C32-4B	Hybrid/EV Battery Cooling System Performance - Over temperature	▪ Hybrid electric vehicle battery pack failure ▪ Air conditioning system fault	▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new hybrid electric vehicle battery pack
P0C33-00	Hybrid Battery Temperature Sensor F Circuit - No sub type information	▪ Hybrid electric vehicle battery pack failure ▪ Battery temperature sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance	▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new hybrid electric vehicle battery pack
P0C34-00	Hybrid Battery Temperature Sensor "F" Circuit Range/Performance - No sub type information	▪ Hybrid electric vehicle battery pack failure ▪ Battery temperature sensor temperature measurement fault	▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new hybrid electric vehicle battery pack
P0C35-14	Hybrid/EV Battery Temperature Sensor "F" Circuit Low - Circuit short to ground or open	▪ Hybrid electric vehicle battery pack failure ▪ Battery temperature sensor signal out of range low	▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new hybrid electric vehicle battery pack
P0C36-12	Hybrid/EV Battery Temperature Sensor "F" Circuit High - Circuit short to	▪ Hybrid electric vehicle battery pack failure ▪ Battery	▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists,

	battery	temperature sensor signal out of range high	install a new hybrid electric vehicle battery pack
P0C43-04	Hybrid Battery Pack Coolant Temperature Sensor "A" Circuit Range/Performance - System internal failures	- Hybrid electric vehicle battery pack failure - Inlet/outlet coolant temperature sensor signal out of range low	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new hybrid electric vehicle battery pack
P0C43-49	Hybrid Battery Pack Coolant Temperature Sensor "A" Circuit Range/Performance - Internal electronic failure	- Hybrid electric vehicle battery pack failure - Inlet/outlet coolant temperature sensor signal out of range high	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new hybrid electric vehicle battery pack
P0C43-62	Hybrid Battery Pack Coolant Temperature Sensor "A" Circuit Range/Performance - Signal compare failure	- Hybrid electric vehicle battery pack failure - Rationality check failed	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new hybrid electric vehicle battery pack
P0C44-11	Hybrid Battery Pack Coolant Temperature Sensor "A" Circuit Low - Circuit short to ground	- Hybrid electric vehicle battery pack failure - Inlet/outlet coolant temperature sensor signal circuit short circuit to ground	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new hybrid electric vehicle battery pack
P0C45-04	Hybrid/EV Battery Pack Coolant Temperature Sensor "A" Circuit High - System internal failures	- Hybrid electric vehicle battery pack failure - Inlet/outlet coolant temperature	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new hybrid electric vehicle battery pack

		sensor signal circuit open circuit	
P0C47-31	Hybrid Battery Pack Coolant Pump Control Circuit / Open - No signal	- Hybrid electric vehicle battery pack cooling system leaking or blocked - Hybrid electric vehicle battery coolant pump failure	- Refer to the relevant section of the workshop manual and test the hybrid electric vehicle battery pack cooling system - Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new hybrid electric vehicle battery coolant pump
P0C47-39	Hybrid Battery Pack Coolant Pump Control Circuit / Open - Incorrect has too few pulses	- Hybrid electric vehicle battery coolant pump control circuit short circuit to ground, short circuit to power, open circuit, high resistance - Hybrid electric vehicle battery coolant pump failure	- Refer to the electrical circuit diagrams and check the hybrid electric vehicle battery coolant pump control circuit for short circuit to ground, short circuit to power, open circuit, high resistance - Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new hybrid electric vehicle battery coolant pump
P0C47-96	Hybrid Battery Pack Coolant Pump Control Circuit / Open - Component internal failure	Hybrid electric vehicle battery coolant pump failure	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new hybrid electric vehicle battery coolant pump
P0C47-98	Hybrid Battery Pack Coolant Pump Control Circuit / Open - Component or system over temperature	Hybrid electric vehicle battery coolant pump over temperature	- Refer to the relevant section of the workshop manual and test the hybrid electric vehicle battery pack cooling system - Remove the hybrid electric

			vehicle battery coolant pump and check for blockages. Check the impeller for free rotation. Reinstall the coolant pump. Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new hybrid electric vehicle battery coolant pump
P0C48-00	Hybrid Battery Pack Coolant Pump Control Circuit Low - No sub type information	- Hybrid electric vehicle battery coolant pump control circuit short circuit to ground, short circuit to power, open circuit, high resistance - Hybrid electric vehicle battery coolant pump failure	- Refer to the electrical circuit diagrams and check the hybrid electric vehicle battery coolant pump control circuit for short circuit to ground, short circuit to power, open circuit, high resistance - Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new hybrid electric vehicle battery coolant pump
P0C49-00	Hybrid Battery Pack Coolant Pump Control Circuit High - No sub type information	- Hybrid electric vehicle battery coolant pump control circuit short circuit to ground, short circuit to power, open circuit, high resistance - Hybrid electric vehicle battery coolant pump failure	- Refer to the electrical circuit diagrams and check the hybrid electric vehicle battery coolant pump control circuit for short circuit to ground, short circuit to power, open circuit, high resistance - Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new hybrid electric vehicle battery coolant pump
P0C4A-1C	Hybrid Battery Pack Coolant Pump Control Performance - Circuit voltage out of range	Hybrid electric vehicle battery pack coolant pump supply circuit short circuit	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new hybrid electric

		to ground, short circuit to power, open circuit, high resistance	vehicle battery pack
P0C4A-42	Hybrid Battery Pack Coolant Pump Control Performance - General memory failure	Hybrid electric vehicle battery pack coolant pump control circuit short circuit to ground, short circuit to power, open circuit, high resistance	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new hybrid electric vehicle battery pack
P0C78-00	Hybrid Battery System Precharge Time Too Long - No sub type information	- High voltage positive circuit short circuit to high voltage negative - Hybrid electric vehicle battery pack failure - Pre-charge contactor stuck open - Negative contactor stuck open	- Refer to the electrical circuit diagrams and check the high voltage positive circuit for short circuit to the high voltage negative circuit - Clear the DTCs and retest. If the fault persists, install a new hybrid electric vehicle battery pack
P0C78-63	Hybrid Battery System Precharge Time Too Long - Circuit/component protection time-out	- High voltage positive circuit short circuit to high voltage negative - Hybrid electric vehicle battery pack failure - Pre-charge contactor stuck open - Negative contactor stuck open	- Refer to the electrical circuit diagrams and check the high voltage positive circuit for short circuit to the high voltage negative circuit - Clear the DTCs and retest. If the fault persists, install a new hybrid electric vehicle battery pack
P0CA6-00	Hybrid Battery Charging Current	Hybrid electric vehicle battery	Using the manufacturer approved diagnostic

	High - No sub type information	pack failure Charge current detected with zero charge power limit reported	system, clear the DTCs and retest. If the fault persists, install a new hybrid electric vehicle battery pack
P0CA7-00	Hybrid Battery Discharging Current High - No sub type information	- Hybrid electric vehicle battery pack failure - Discharge current detected with zero charge power limit reported	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new hybrid electric vehicle battery pack
P0D34-00	Hybrid/EV Battery System Current - No sub type information	- Hybrid electric vehicle battery pack failure - Reserve over-heating capacity negative	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new hybrid electric vehicle battery pack
P0D37-00	Hybrid/EV Battery System Current High - No sub type information	- Hybrid electric vehicle battery pack failure - Reserve lithium plating capacity negative	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new hybrid electric vehicle battery pack
P0D6C-00	A/C Compressor Motor Voltage Sensor Circuit Intermittent/Erratic - No sub type information	High voltage circuit to the electric air conditioning compressor open circuit	NOTE: Rectify high voltage interlock loop DTCs first Refer to the electrical circuit diagrams and check the high voltage circuit to the electric air conditioning compressor for open circuit

P0DA8-00	Hybrid/EV Battery Voltage/Drive Motor "A" Inverter Voltage Correlation - No sub type information	High voltage circuit to electric power inverter converter open circuit	NOTE: Rectify high voltage interlock loop DTCs first Refer to the electrical circuit diagrams and check the high voltage circuit to electric power inverter converter for open circuit
P0DA9-00	Hybrid/EV Battery Voltage/Drive Motor "B" Inverter Voltage Correlation - No sub type information	High voltage circuit to electric power inverter converter open circuit	NOTE: Rectify high voltage interlock loop DTCs first Refer to the electrical circuit diagrams and check the high voltage circuit to electric power inverter converter for open circuit
P0DAF-00	Hybrid/EV Battery Cell Balancing Circuit "A" Range/Performance - No sub type information	- Hybrid electric vehicle battery pack failure - Offset balance address check failed	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new hybrid electric vehicle battery pack
P0DB3-00	Hybrid/EV Battery Cell Balancing Circuit "B" Range/Performance - No sub type information	- Hybrid electric vehicle battery pack failure - Offset balance address check failed	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new hybrid electric vehicle battery pack
P0DB7-00	Hybrid/EV Battery Cell Balancing Circuit "C" Range/Performance - No sub type information	- Hybrid electric vehicle battery pack failure - Offset balance address check failed	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new hybrid electric vehicle battery pack

P0DBB-00	Hybrid/EV Battery Cell Balancing Circuit "D" Range/Performance - No sub type information	- Hybrid electric vehicle battery pack failure - Offset balance address check failed	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new hybrid electric vehicle battery pack
P0DBF-00	Hybrid/EV Battery Cell Balancing Circuit "E" Range/Performance - No sub type information	- Hybrid electric vehicle battery pack failure - Offset balance address check failed	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new hybrid electric vehicle battery pack
P0DC3-00	Hybrid/EV Battery Cell Balancing Circuit "F" Range/Performance - No sub type information	- Hybrid electric vehicle battery pack failure - Offset balance address check failed	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new hybrid electric vehicle battery pack
P0DE6-00	Hybrid/EV Battery Pack Cell Voltage Low - No sub type information	- Hybrid electric vehicle battery pack failure - Cell voltage low	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new hybrid electric vehicle battery pack
P0DE6-16	Hybrid/EV Battery Pack Cell Voltage Low - Circuit voltage below threshold	- Hybrid electric vehicle battery pack failure - Cell voltage below minimum limit	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new hybrid electric vehicle battery pack
P0DE7-00	Hybrid/EV Battery Pack Cell Voltage High - No sub type information	- Hybrid electric vehicle battery pack failure - Cell voltage high	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new hybrid electric vehicle battery pack
P0DE7-17	Hybrid/EV Battery Pack Cell Voltage High - Circuit voltage above threshold	- Hybrid electric vehicle battery pack failure - Cell voltage	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists,

		above maximum limit	install a new hybrid electric vehicle battery pack
P0E3B-7B	Hybrid/EV Battery Coolant Pump Overspeed - Low fluid level	Hybrid electric vehicle battery pack cooling system leaking	Refer to the relevant section of the workshop manual and test the hybrid electric vehicle battery pack cooling system
P0E3C-97	Hybrid/EV Battery Coolant Pump Underspeed - Component or system operation obstructed or blocked	Hybrid electric vehicle battery coolant pump blocked	Remove the hybrid electric vehicle battery coolant pump and check for blockages. Check the impeller for free rotation. Reinstall the coolant pump. Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new hybrid electric vehicle battery coolant pump
P0E74-00	Hybrid/EV Battery Voltage Internal Isolation Fault - No sub type information	- Hybrid electric vehicle battery pack failure - High voltage - low isolation resistance to ground inside hybrid electric vehicle battery pack	NOTE: If DTC P1A21-00 is also present, rectify that DTC first Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new hybrid electric vehicle battery pack
P0ECA-00	Hybrid/EV Battery Temperature Sensor Correlation Error - No sub type information	- Hybrid electric vehicle battery pack failure - Battery multiple temperature sensor faults	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new hybrid electric vehicle battery pack
P1608-	Watchdog	Hybrid electric	Using the manufacturer

49	Malfunction - Internal electronic failure	vehicle battery pack failure ▪ Secondary safety monitor failure	approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new hybrid electric vehicle battery pack
P1A0B-04	Hybrid Battery Power Limit Exceeded - System internal failures	▪ Hybrid electric vehicle battery pack failure ▪ Invalid cell voltage detected	▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new hybrid electric vehicle battery pack
P1A0B-49	Hybrid Battery Power Limit Exceeded - Internal electronic failure	▪ Hybrid electric vehicle battery pack failure ▪ Invalid cell voltage detected	▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new hybrid electric vehicle battery pack
P1A1E-00	Hybrid/EV Battery Contactor Control Circuit/Open - No sub type information	▪ Hybrid electric vehicle battery pack failure ▪ Main positive contactor driver high side circuit short circuit to ground, open circuit	▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new hybrid electric vehicle battery pack
P1A1F-00	Hybrid/EV Battery Contactor Control Circuit Range/Performance - No sub type information	▪ Hybrid electric vehicle battery pack failure ▪ Main positive contactor driver high side feedback circuit voltage above threshold	▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new hybrid electric vehicle battery pack
P1A1F-16	Hybrid/EV Battery Contactor Control Circuit Range/Performance - Circuit voltage below threshold	▪ Hybrid electric vehicle battery pack failure ▪ Contactor high-side driver input voltage out of	▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new hybrid electric vehicle battery pack

		range low	
P1A1F-17	Hybrid/EV Battery Contactor Control Circuit Range/Performance - Circuit voltage above threshold	- Hybrid electric vehicle battery pack failure - Contactor high-side driver input voltage out of range high	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new hybrid electric vehicle battery pack
P1A20-00	Hybrid/EV Battery Contactor Control Circuit High - No sub type information	- Hybrid electric vehicle battery pack failure - Main positive contactor driver high side circuit short circuit to power	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new hybrid electric vehicle battery pack
P1A21-00	Hybrid/EV Battery Water Intrusion Detected - No sub type information	- Hybrid electric vehicle battery pack failure - Water ingress	NOTE: Damaged high voltage cables can allow water into high voltage components - Perform a visual inspection of all high voltage component breather pipes (hybrid electric vehicle battery pack, electric power inverter converter and transmission) for damage and water ingress - Install a new hybrid electric vehicle battery pack. Incline original battery pack to allow any water to drain - Check the hybrid electric vehicle battery pack breather pipe and cables and connectors for damage
U0001-	High Speed CAN	Missing message	Using the manufacturer

00	Communication Bus - No sub type information	from another control module via the high speed CAN bus (hybrid)	approved diagnostic system, check the snapshot data to determine the missing message source control module. Check the relevant control module for related DTCs and refer to the relevant DTC index
U0073- 00	Control Module Communication Bus "A" Off - No sub type information	High speed CAN bus (hybrid) circuit short circuit to ground, short circuit to power, open circuit, high resistance	Using the manufacturer approved diagnostic system, perform a CAN network integrity test. Refer to the electrical circuit diagrams and check the high speed CAN bus (hybrid) circuit for short circuit to ground, short circuit to power, open circuit, high resistance
U0100- 00	Lost Communication With ECM/PCM "A" - No sub type information	Missing message from another control module via the high speed CAN bus (hybrid)	Using the manufacturer approved diagnostic system, check the snapshot data to determine the missing message source control module. Check the relevant control module for related DTCs and refer to the relevant DTC index
U0110- 00	Lost Communication With Drive Motor Control Module A - No sub type information	Missing message from another control module via the high speed CAN bus (hybrid)	Using the manufacturer approved diagnostic system, check the snapshot data to determine the missing message source control module. Check the relevant control module for related DTCs and refer to the relevant DTC index
U0146- 00	Lost Communication With Gateway "A" - No sub type information	Missing message from another control module via the high speed CAN bus (hybrid)	Using the manufacturer approved diagnostic system, check the snapshot data to determine the missing message source control module. Check the relevant control module for related DTCs and refer to the relevant DTC index

U0147-00	Lost Communication With Gateway "B" - No sub type information	Missing message from another control module via the high speed CAN bus (hybrid)	Using the manufacturer approved diagnostic system, check the snapshot data to determine the missing message source control module. Check the relevant control module for related DTCs and refer to the relevant DTC index
U0148-00	Lost Communication With Gateway "C" - No sub type information	Missing message from another control module via the high speed CAN bus (hybrid)	Using the manufacturer approved diagnostic system, check the snapshot data to determine the missing message source control module. Check the relevant control module for related DTCs and refer to the relevant DTC index
U0292-00	Lost Communication With Drive Motor Control Module "B" - No sub type information	Missing message from another control module via the high speed CAN bus (hybrid)	Using the manufacturer approved diagnostic system, check the snapshot data to determine the missing message source control module. Check the relevant control module for related DTCs and refer to the relevant DTC index
U0300-00	Internal Control Module Software Incompatibility - No sub type information	- High speed CAN bus (hybrid) circuit short circuit to ground, short circuit to power, open circuit, high resistance - Car configuration file mismatch with vehicle specification - Incorrect battery energy control module installed	- Using the manufacturer approved diagnostic system, perform a CAN network integrity test. Refer to the electrical circuit diagrams and check the high speed CAN bus circuit for short circuit to ground, short circuit to power, open circuit, high resistance - Using the manufacturer approved diagnostic system, check and update the car configuration file as necessary - Install a new hybrid electric vehicle battery pack as necessary

U0401-00	Invalid Data Received from ECM/PCM A - No sub type information	Invalid data received from another control module via the high speed CAN bus (hybrid)	Using the manufacturer approved diagnostic system, check the snapshot data to determine the invalid data source control module. Check the relevant control module for related DTCs and refer to the relevant DTC index
U0594-66	Invalid Data Received From Hybrid Powertrain Control Module - Signal has too many transitions/events	Ignition cycled on/off more than 3 times within 20 seconds	NOTE: If the ignition is cycled on/off more than 3 times within 20 seconds, this DTC will set and the hybrid system will be inhibited for a period of 2 minutes. Full functionality is then restored No action required (this DTC is for information only)
U3001-00	Control Module Improper Shutdown - No sub type information	- Battery energy control module power or ground (low voltage) circuit open circuit, high resistance - Battery fault (low voltage) - Charging system fault (low voltage)	- Refer to the electrical circuit diagrams and check the battery energy control module power and ground (low voltage) circuits for open circuit, high resistance - Refer to the relevant section of the workshop manual and test the battery - Using the manufacturer approved diagnostic system, check the gateway module for related DTCs and refer to the relevant DTC index
U3003-12	Battery Voltage - Circuit short to battery	Charging system fault (low voltage)	Using the manufacturer approved diagnostic system, check the gateway

			module for related DTCs and refer to the relevant DTC index
U3003-14	Battery Voltage - Circuit short to ground or open	▪ Battery energy control module power or ground (low voltage) circuit open circuit, high resistance ▪ Battery fault (low voltage) ▪ Charging system fault (low voltage)	▪ Refer to the electrical circuit diagrams and check the battery energy control module power and ground (low voltage) circuits for open circuit, high resistance ▪ Refer to the relevant section of the workshop manual and test the primary battery ▪ Using the manufacturer approved diagnostic system, check the gateway module for related DTCs and refer to the relevant DTC index
U3003-16	Battery Voltage - Circuit voltage below threshold	▪ Battery energy control module power or ground (low voltage) circuit open circuit, high resistance ▪ Battery fault (low voltage) ▪ Charging system fault (low voltage)	▪ Refer to the electrical circuit diagrams and check the battery energy control module power and ground (low voltage) circuits for open circuit, high resistance ▪ Refer to the relevant section of the workshop manual and test the primary battery ▪ Using the manufacturer approved diagnostic system, check the gateway module for related DTCs and refer to the relevant DTC index
U3003-17	Battery Voltage - Circuit voltage above threshold	▪ Charging system fault (low voltage)	▪ Using the manufacturer approved diagnostic system, check the gateway module for related DTCs and refer to the relevant DTC index
U3006-00	Control Module Input Power "A" - No sub	▪ Battery energy control module	▪ Refer to the electrical circuit diagrams and check the

	type information	power or ground (low voltage) circuit open circuit, high resistance ▪ Battery fault (low voltage) ▪ Charging system fault (low voltage)	battery energy control module power and ground (low voltage) circuits for open circuit, high resistance ▪ Refer to the relevant section of the workshop manual and test the primary battery ▪ Using the manufacturer approved diagnostic system, check the gateway module for related DTCs and refer to the relevant DTC index
U3017-00	Control Module Timer/Clock Performance - No sub type information	▪ Hybrid electric vehicle battery pack failure ▪ System clock failure	▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new hybrid electric vehicle battery pack